Comprehensive Road Scene Understanding for Autonomous Driving: from Mask-Architecture Segmentation to Anomaly Segmentation

Group 29 - Project report
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Abstract

Semantic segmentation is a key component of autonomous-driving perception, but standard models are usually trained in a closed-set setting and may fail when unknown objects appear on the road. In this work, we study the performance of a mask-transformer segmenter in anomaly segmentation as a post-hoc problem. We compare a pixel-based ERFNet baseline with several Encoder-only Mask Transformer (EoMT) variants, including COCO-trained, Cityscapes-trained, and progressively fine-tuned checkpoints. First, we evaluate semantic segmentation on Cityscapes dataset, showing that the Cityscapes-trained EoMT is the strongest segmenter, while staged fine-tuning reduces the gap between COCO pretraining and the road scene domain. Then, we evaluate anomaly segmentation on five road anomaly benchmarks using Maximum Softmax Probability (MSP), Maximum Logit Score (MaxLogit), Maximum Entropy, Rejected by All (RbA) and temperature-scaled variants. The results show that domain-adapted mask-based models clearly outperform the pixel baseline for anomaly segmentation. They also show that the best semantic segmenter is not always the best anomaly segmenter: the fine-tuned EoMT obtained in the last stage of the progressive training achieves on average better anomaly performance. Finally, we show that the anomaly score and calibration strategy strongly affect the results, highlighting the importance of combining domain adaptation with suitable post-hoc scoring for open-world road-scene understanding. Our code is available at: `https://github.com/ChiaraApolito/MaskArchitectureAnomaly_CourseProject.git`

1. Introduction

Image segmentation and anomaly segmentation are closely related tasks that together enable robust scene understanding in a wide range of applications. Their combination is particularly relevant in autonomous driving, medical imaging [17], industrial inspection [1], and security monitoring [8], where correctly identifying both known and unknown objects is essential and may be safety critical. In this work, we focus on the autonomous driving domain, where the perception task requires a good trade-off between high quality and computational resources.

Segmentation for autonomous driving. Early solutions based on Convolutional Neural Networks (CNNs), such as ERFNet [16], demonstrated impressive capabilities in addressing the task of real-time semantic segmentation, that assigns class label to each pixel. Instance segmentation instead detects and delineates individual instances of object. Panoptic segmentation [12] combines both semantic segmentation and instance segmentation. In panoptic segmentation, *things* classes (e.g. cars or pedestrians) and *stuff* classes (amorphous and uncountable regions such as road or sky) are processed in a unified manner to provide a global view of the scene. Recent research has moved from pure per-pixel classification toward mask-based formulations. MaskFormer [5] showed that segmentation can be treated as mask classification rather than independent pixel classification, and Mask2Former [6] further improved this paradigm through masked attention and a unified architecture for semantic, instance, and panoptic segmentation. More recently, DINOv2 [15] demonstrated the strength of large-scale self-supervised visual representations, while EoMT [11] showed that a plain Vision Transformer can be effectively repurposed for image segmentation with a simpler and faster architecture.

Anomaly segmentation. However, most segmentation models are trained under a closed-set assumption, where every pixel is expected to belong to a known training class. This is unsafe in real-world driving scenarios, where unknown or out-of-distribution objects may appear unexpectedly — objects that the model has never seen during training and therefore cannot correctly classify. Benchmark datasets such as Fishyscapes [2] and SegmentMeIfYouCan [3] were introduced specifically to measure the ability of segmentation models to identify such unknown objects on the road, providing evaluation sets that contain anomalies absent from standard training datasets like Cityscapes [7]. Several post-hoc methods have been proposed to address this problem without retraining the model.

In this paper, we study how to turn a mask-transformer semantic segmenter (EoMT) into a reliable post-hoc out-of-distribution segmenter for road scenes, and how architecture, domain adaptation and score calibration affect this ability. Specifically, we first fine-tune a COCO-pretrained EoMT model on Cityscapes dataset in progressive stages, analyzing the impact of each stage on semantic segmentation performance. We then study road anomaly segmentation by comparing a pixel-based approach (ERFNet) and multiple EoMT-based checkpoints, evaluating several post-hoc anomaly scoring strategies including MSP [9], MaxLogit [10], Max Entropy [4] and RbA [14] - which exploits the query-based structure of mask architectures - and analyzing how fine-tuning affects anomaly segmentation performance.

2. Methodology

Our goal is to use a mask-transformer semantic segmenter, EoMT, as a post-hoc out-of-distribution detector for road scenes. We study the comparative performance in both semantic and anomaly segmentation for several models. A Cityscapes-trained model is expected to work well on known road-scene classes, while a COCO-pretrained model may retain broader visual knowledge acquired from the 133 object categories of the COCO dataset [13] spanning diverse domains and scene types. After fine-tuning on Cityscapes, the COCO-pretrained model can combine road-scene adaptation with useful information from COCO.

The methodology has three main steps. First, we evaluate different pretrained models for semantic segmentation. Second, we fine-tune the COCO-trained EoMT model on Cityscapes. Third, we evaluate all models for post-hoc anomaly segmentation.

We compare two types of segmentation models. ERFNet is used as an efficient convolutional baseline for pixel-wise semantic segmentation. EoMT is used as a mask-based model built on Vision Transformer features.

2.1. Semantic Segmentation Evaluation: mapping strategy

We first evaluate two pretrained EoMT models on the Cityscapes validation set. One model is trained on Cityscapes for semantic segmentation, while the other is trained on COCO for panoptic segmentation. The Cityscapes-trained model can be evaluated directly on the 19 Cityscapes classes. The COCO-trained model uses a different label space, so its predictions must be mapped to Cityscapes classes.

Some COCO classes have a direct match in Cityscapes, but other cases are many-to-one. For example, several COCO building-related classes are mapped to the Cityscapes *building* class, and several greenery-related classes are mapped to *vegetation*. When more COCO classes map to the same Cityscapes class, we take the maximum logit among them instead of summing the logits. This avoids giving an unfair advantage to Cityscapes classes that receive many COCO classes. Since *pole*, *traffic sign*, and *rider* do not have a clear COCO match, we include their pixels in the *void* class. Therefore, we report a 16-classes evaluation that excludes them.

2.2. Fine-tuning Procedure

To reduce the gap between COCO panoptic pretraining and Cityscapes semantic segmentation, we fine-tune the COCO-trained EoMT model on the Cityscapes training set. Because of GPU memory limits, all fine-tuning experiments use a reduced input resolution of 512×512 and Automatic Mixed Precision (AMP) - 16-mixed AMP in Baseline and the more stable version bfloat16 in other experiments. We use a progressive fine-tuning strategy, where each stage trains more parameters than the previous one:

1. **Baseline**: class head only, using cached query features for efficiency (50 epochs);
2. **Stage 0**: class head and mask head (10 epochs);
3. **Stage 1**: full prediction head with the DINOv2 backbone frozen (15 epochs);
4. **Stage 2**: full prediction head with the last two DINOv2 blocks unfrozen (40 epochs).

The baseline and Stage 0 are independent experiments starting from the same COCO-trained checkpoint, while Stage 1 and Stage 2 are initialized from the checkpoint of the previous stage. This progressive-unfreezing strategy adapts the new trainable layers while keeping the pretrained features stable.

Each resulting model is compared with the initial COCO-trained model and with the original Cityscapes-trained reference model, following the protocol in Section 2.1. The fine-tuned Stage 2 model is then selected for the anomaly segmentation evaluation.

2.3. Anomaly Segmentation Evaluation

Finally, we evaluate anomaly segmentation as a post-hoc scoring problem. The models are not trained with an explicit unknown class. Instead, anomaly maps are computed from the standard segmentation output using confidence-based or rejection-based scores. We test MSP, MaxLogit, Maximum Entropy for ERFNet and the EoMT models, while RbA for EoMT only. For EoMT, MaxLogit refers to a MaxLogit-like score computed from the reconstructed pixel-level semantic scores. We use a RbA-style score adapted to the mask-query output: class-wise semantic

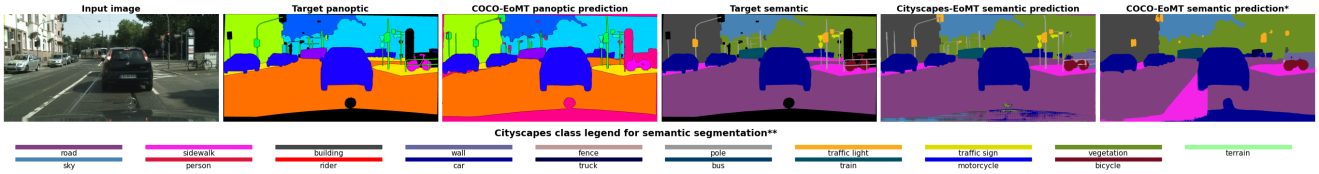

Figure 1. **Qualitative comparison of 2 EoMT models.** From left to right: input image nr. 8, ground truth (GT) panoptic, panoptic prediction from COCO-trained model on 133 classes, GT semantic, semantic prediction from Cityscapes-trained model on 19 classes and semantic prediction from COCO-trained model, (*) shown after mapping of the class space. (**) The class legend is not applicable to panoptic segmentation. The comparison between the semantic prediction of the models shows how the Cityscapes model is better aligned to the ground truth (e.g. part of the road is wrongly classified as sidewalk from the COCO-trained model).

scores are reconstructed by combining query masks and query class probabilities, then the anomaly score is obtained by measuring how weakly each pixel is accepted by the known classes. Temperature-scaled variants are reported for MSP applied to EoMT finetuned Stage 2 model.
We evaluate the models on five road anomaly benchmarks. These datasets differ in scene type, anomaly definition, and object size testing different types of road-scene anomalies. RoadAnomaly21 and RoadAnomaly consider as anomalous any object category that is not part of the 19 Cityscapes classes. They include different environments and anomalies that can cover up to 40% of the image. RoadObstacle21 focuses on safety-critical objects on the road: any object in front of the vehicle is considered dangerous, even if it belongs to a known class. Fishyscapes Lost & Found uses real driving sequences with small anomalous objects in urban scenes. Fishyscapes Static overlays anomalous objects on Cityscapes validation images, giving a controlled synthetic benchmark.

2.4. Evaluation Metrics

We use task-specific metrics. For semantic segmentation, we report mean Intersection over Union (mIoU), computed as the average IoU over the evaluated classes and pixel accuracy. For anomaly segmentation, we report the Area under the Precision-Recall Curve (AuPRC), a key metric as anomaly datasets are highly imbalanced, and the False Positive Rate at 95% True Positive Rate (FPR95) that measures how many normal pixels are wrongly classified as anomalous when the detector finds 95% of the true anomalies.

3. Experimental Results

We present the experimental results following the three-stage methodology described in Section 2.

3.1. Cross-domain comparison of EoMT

Table 1 reports the quantitative comparison between the two pretrained EoMT checkpoints on the Cityscapes validation set, over the 16 matching classes. The Cityscapes-trained model obtains clearly better results than the COCO-trained

Model	mIoU (%)	pixel acc. (%)
EoMT, COCO $\rightarrow$ Cityscapes	67.01	93.39
EoMT, Cityscapes	**83.74**	**97.04**

Table 1. **Quantitative results of the semantic segmentation comparison** between COCO-pretrained and Cityscapes-pretrained EOMT models on the Cityscapes validation set for 16 matching classes.

model, with about 17 more mIoU points and almost 4 more pixel accuracy points, as also shown qualitatively in Fig. 1 .

3.2. Fine-tuning on Cityscapes training dataset

Table 2 reports the results of our progressive fine-tuning, compared against the two initial pre-trained models - EoMT COCO and EoMT Cityscapes. The mIoU of the COCO-trained fine-tuned model improves at each stage, from 67.01% of the initial checkpoint on the 16-classes protocol to 77.15% of the last stage, representing a gain of over 10 mIoU points. Nevertheless, the fine-tuned model does not reach the 83.74% mIoU of the model trained directly on Cityscapes.

Model	**Trainable parts**	**mIoU (%)**	
		16-cl.	**19-cl.**
finetuned Baseline	Class head only	71.14	62.20
finetuned Stage 0	Class head + mask head	74.73	66.64
finetuned Stage 1	Full head (frozen DINO)	76.34	70.57
finetuned Stage 2	Full head + 2 last DINO blocks	**77.15**	72.98
EoMT COCO		67.01	–
EoMT Cityscapes		**83.74**	81.68

Table 2. **Fine-tuning set-up and results.** The trainable parts of the model are increased at each step of the fine-tuning from *finetuned Baseline* to *finetuned Stage 2*. EoMT COCO and EoMT Cityscapes initial models are reported as a reference. MIoU is evaluated over 16 and 19 classes.

Model	mIoU %	Method	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓	AuPRC↑	FPR95↓
ERFNet	75.10	MSP	29.08	62.56	2.71	65.18	1.75	50.66	7.47	41.84	12.42	82.58
		MaxLogit	**38.30**	**59.37**	**4.63**	**48.44**	**3.30**	**45.53**	**9.50**	**40.30**	**15.58**	**73.28**
		Max Entropy	30.96	62.68	3.05	65.88	2.58	50.21	8.84	41.55	12.67	82.76
EoMT COCO	67.01	MSP	21.06	89.54	7.40	100	0.73	52.54	4.46	99.76	11.39	94.12
		MaxLogit	**39.65**	72.66	2.62	99.99	1.28	95.47	4.41	99.74	**22.92**	97.10
		Max Entropy	23.30	89.43	13.96	100	**1.28**	**44.38**	**4.85**	99.72	17.79	93.53
		RbA	23.21	**64.78**	**52.62**	99.99	0.18	97.12	1.87	**99.06**	16.46	**82.83**
EoMT Cityscapes	83.74	MSP	54.22	33.03	71.47	40.76	2.53	44.25	10.66	46.25	43.71	41.89
		MaxLogit	**67.85**	42.16	**88.69**	**13.99**	15.90	19.31	48.69	**27.72**	**69.47**	**14.21**
		Max Entropy	63.69	**30.30**	75.53	38.48	3.65	45.83	12.80	46.09	49.45	41.63
		RbA	66.36	76.19	85.22	99.95	**16.52**	**14.73**	**49.90**	34.40	69.08	17.66
EoMT finetuned stage2	77.15	MSP (T=1)	65.02	**29.23**	79.91	4.49	9.52	59.63	55.21	27.59	66.55	37.25
		MaxLogit	57.12	90.30	77.03	**2.37**	**37.87**	**20.65**	**86.52**	**7.46**	61.26	35.30
		Max Entropy	**76.28**	29.38	**85.73**	3.63	16.35	59.92	69.88	26.57	**75.66**	36.57
		RbA	64.91	35.80	81.33	4.39	33.72	22.18	83.18	10.35	71.91	**26.65**
		MSP (T=0.5)	44.18	32.02	55.60	8.78	1.34	43.02	19.33	42.40	41.23	46.75
		MSP (T=0.75)	53.92	30.05	71.62	6.32	4.91	46.23	38.07	33.46	57.27	40.88
		MSP (T=1.1)	67.48	29.77	80.99	3.90	11.09	63.33	59.42	25.67	68.51	35.87

Table 3. **Anomaly segmentation scores** of ERFNet and EoMT-based models across different post-hoc methods and datasets. MIoU is evaluated with 16-classes protocol for all models allowing fair comparison with COCO-trained one. The best temperature scaled scores are underline for MSP method applied on the finetuned model in each dataset. Bolded values are the best results of AuPRC and FPR95 for each combination of model and anomaly segmentation dataset.

3.3. Anomaly Segmentation

Table 3 reports the outcome of the post-hoc anomaly segmentation evaluation in terms of AuPRC and FPR95, for each combination of model and scoring method across the five benchmarks. The Cityscapes-trained model and our Stage 2 fine-tuned model outperform both EoMT COCO and ERFNet on all benchmarks. Overall, the Stage 2 model achieves the best AuPRC on four out of five datasets and the best FPR95 on three of them. In the case of ERFNet, MaxLogit consistently achieves the best AuPRC across all benchmarks. Instead, a superior method cannot be determined across the EoMT-based models: the best-performing method varies depending on both the model and the dataset, with Max Entropy, MaxLogit, and RbA each outperforming the others in different settings. Figure 2 shows anomaly segmentation performance of our Stage 2 finetuned variant and the initial Cityscapes model on two sample images.

3.4. Temperature scaling

We also test temperature scaling with $T \in \{0.5, 0.75, 1.0, 1.1\}$. Table 3 reports the results for MSP method applied to the Stage 2 fine-tuned model. As underlined in the table, this calibration shows that MSP benefits from higher temperature, with the results improving at $T{=}1.1$ with respect to the baseline $T{=}1.0$. Only in the case of Fishyscapes Lost & Found, the best results is achieved at a lower temperature, however, this is the only dataset where the fine-tuned Stage 2 model does not perform well.

4. Discussion

4.1. Result analysis

Impact of task and domain shift. The comparison of the two initial EoMT checkpoints (Tab. 1, Fig. 1) shows that the Cityscapes-trained version substantially outperforms the COCO-trained model. This difference in performance can be attributed to the *task gap* and a *domain gap* between the two checkpoints. The COCO-pretrained model is optimized for panoptic segmentation task on a wide variety of scenes and object categories, whereas Cityscapes-pretrained model targets directly semantic segmentation on road scenes.

Why the mask architecture works better. The pixel-based ERFNet baseline shows a weaker performance than EoMT Cityscapes on all anomaly benchmarks (Tab. 3). This result can be explained by the fundamental difference between the two architectures: ERFNet applies a per-pixel softmax over the known training classes, thus forcing high confidence on pixels with anomalies. Instead, EoMT's query-based formulation can be exploited for uncertainty estimation, since regions that are weakly explained by all known-class queries can receive higher anomaly scores.

The best segmenter is not always the best anomaly segmenter. The initial Cityscapes model is the strongest se-

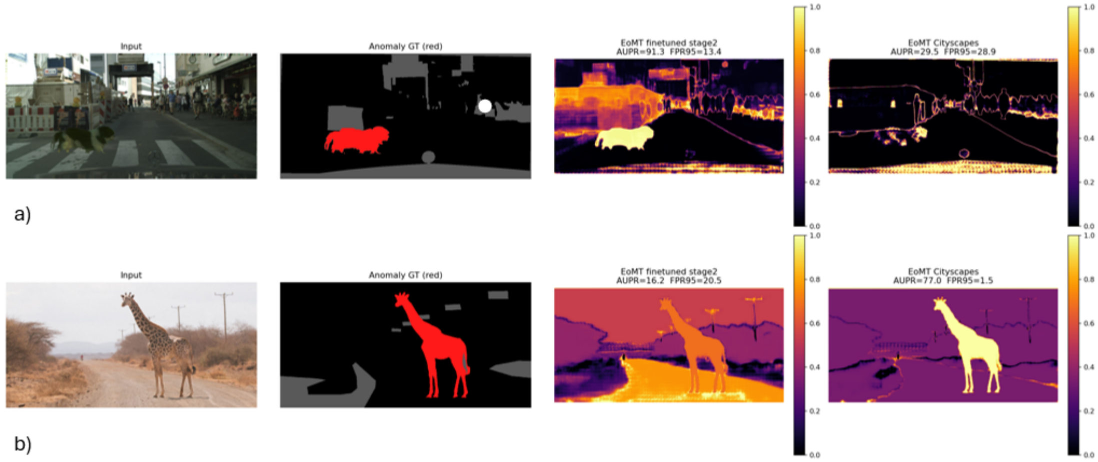

Figure 2. **Qualitative anomaly maps**. **(a)** Fishyscapes Static with MSP, success case: the Stage 2 fine-tuned model gives high anomaly scores mainly to the small unknown object (AuPR 91.3), while the Cityscapes model reacts much less (29.5). **(b)** RoadAnomaly21 with RbA, failure case: the Stage 2 fine-tuned model assigns high scores also to the unfamiliar dirt road, producing false positives. Its AuPR decreases to 16.2, while the Cityscapes model reaches 77.0 by keeping the road as a known region and isolating the animal.

mantic segmenter (Tab. 2), but the COCO-pretrained model fine-tuned up to Stage 2 performs better in anomaly segmentation (Tab. 3). Our hypothesis is that, after fine-tuning, the COCO-pretrained model retains part of the broad knowledge coming from the pretraining, which may help in detecting when an object does not belong to any of the reduced Cityscapes semantic space. Progressive fine-tuning improves the semantic segmentation performance of the COCO-pretrained model, with the most significant gain occuring when the mask head is unlocked. The remaining gap with the Cityscapes reference can be partially attributed to the limited fine-tuning. However, a deeper adaptation to Cityscapes may come at the cost of reduced anomaly segmentation performance.

No single post-hoc score is the best in all cases. MaxLogit and RbA depend strongly on the absolute confidence of the model. They work well when the model is well calibrated, but they can produce many false positives when the model gives uncertain scores on normal regions. This is especially relevant for RbA: even though it is naturally suited to mask architectures, it assumes that anomalous pixels are rejected by all known-class queries. Therefore, difficult normal regions that are weakly accepted by the model, such as unfamiliar road surfaces or ambiguous boundaries, can be incorrectly assigned high anomaly scores. At the same time, anomalous objects that are still confidently covered by a query may not be strongly rejected. MSP and MaxEntropy instead depend more on the shape of the output distribution. For this reason, they can be more stable for the Stage 2 model on generic anomaly datasets. The best score therefore depends on both the benchmark and the adaptation strategy. Temperature scaling reduces part of this problem (Tab. 3). A softer distribution, given by T=1.1, positively impacts MSP by increasing the separation between well-explained normal pixels and uncertain anomalous regions.

4.2. Methodological critique

Several limitations should be considered when interpreting the results. (i) The fine-tuning was performed under limited computational resources with reduced resolution and some blocks excluded from the training. (ii) For each benchmark dataset, we report several post-hoc scores. In a real application case, these choices should be fixed before deployment. (iii) Anomaly segmentation is evaluated as a post-hoc scoring problem only, with the models not being specifically trained on this task. Therefore, further development of this work should include: full-resolution, deeper fine-tuning, training on anomaly segmentation task and score selection.

5. Conclusion

We evaluated EoMT for both semantic segmentation and post-hoc anomaly segmentation in road scenes. The initial Cityscapes-trained model reaches the highest mIoU on semantic segmentation on the shared 16-class space. The fine-tuning under limited resources improves the initial performance of the COCO-trained model without outperforming Cityscapes-trained. Interestingly, the results show that the best semantic segmenter is not always the best anomaly segmenter, with the Stage 2 fine-tuned model often obtaining the best performance on the considered anomaly benchmarks. The results also show that the choice of the anomlay score method matters and temperature scaling can further improve calibration.

Acknowledgments

We assure that this report is not created by generative AI-based tools.

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- ChatGPT GPT-5.5 by OpenAI,
- Claude Sonnet 4.6 by Anthropic.

The context and examples of tasks where the above mentioned tools were used are listed below:
- support in debugging,
- quicker *README* writing,
- code structuring and cleaning,
- generation of output artifacts such as tables and figures.

Cross-platform usage was intentional to reduce dependence from a single platform. All AI-assisted contributions were always reviewed and validated by the authors, by applying independent engineering judgment.

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